

Smit Chaudhary

SOFTWARE ENGINEER · QUANTUM ALGORITHMS DEVELOPER · QUANTUM PHYSICIST

Rotterdam The Netherlands

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Experience

Quantum Algorithms Developer

Amsterdam, The Netherlands

PASQAL

2022 - Present

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Quantum Computing Intern

California, USA

MENTEN AI

May 2022 - Sept 2022

- Developed a novel fully quantum generative adversarial network (QGAN) architecture for discrete and binary data generation
- Investigated and implemented quantum circuit features such as noise reuploading in the generator and auxiliary qubits in the discriminator to enhance expressivity
- Demonstrated the model's performance on synthetic datasets and low-energy states of Ising models, showing successful data reproduction and generalization capabilities

Education

Technische Universiteit Delft

Delft, The Netherlands

MASTER OF SCIENCE IN APPLIED PHYSICS

August 2020 - July 2022

- **Thesis:** Quantum Walk Based Algorithms for Qubit Mapping (**Supervisor:** Prof. Sebastian Feld)
- **Relevant Coursework:** Applied Quantum Algorithms, Quantum Information, Quantum Computing Architecture
- **Research Focus:** Near term and variational quantum algorithms

Indian Institute of Technology, Kanpur

Kanpur, India

BACHELOR OF SCIENCE IN PHYSICS

July 2016 - May 2020

- **Relevant Coursework:** Quantum Computing, Quantum Field Theory, Statistical Mechanics, Probability and Statistics, Optics
- **Undergraduate Project:** Bohmian Mechanics and Quantum Information (**Supervisor:** Prof. Kaushik Bhattacharya)
- **Research Experience:** Summer Research at IISER Kolkata on Quantum Machine Learning (**Supervisor:** Prof. Prasanta Panigrahi)

Selected Publications

Solving Fluid Dynamics Equations with Differentiable Quantum Circuits

S. Chaudhary, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile

Proceedings of the 35th Parallel CFD International Conference 2024, May 2025

Quantum Circuit Training with Growth-Based Architectures

C. Duffy, S. Chaudhary, G. V. Velikova

arXiv preprint, Nov 2024

Towards a scalable discrete quantum generative adversarial neural network

S. Chaudhary, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda

Quantum Science and Technology, April 2023

Skills

Programming

Python, C/C++, Rust, Verilog

Languages

English, Hindi, Gujarati

Libraries & Utilities

Git, PyTorch, PennyLane, Tensorflow, Jax, Cirq, Qiskit